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PART **2**

LINEAR ALGEBRA

VECTOR SPACE

Definition: Let $(F, +, \cdot)$ be a field. The elements of F will be called scalars. Let V be a non empty set whose elements will be called vectors. Then V is a Vector space over the field F , if

1. Internal composition in V called addition of vectors is defined. It is denoted by $+$. also for this composition V is an abelian group i.e.,
 - (i)
 - (ii)
 - (iii)
 - (iv)
 - (v)
2. There is an external composition in V over F called scalar multiplication. $a\alpha \in V$ for all $a \in F$ and for all $\alpha \in V$.
3. The two composition i.e. Scalar multiplication and vector addition satisfy the following postulates:-
 - (i) $a(\alpha + \beta) = a\alpha + a\beta$. $\forall a \in F$ and $\forall \alpha \in V$
 - (ii) $(a + b)\alpha = a\alpha + b\alpha$
 - (iii) $(ab)\alpha = a(b\alpha)$
 - (iv) $1\alpha = \alpha \forall \alpha \in V$ and 1 is the unity element of the field F .

When V is a vector space over the field F , we shall say the $V(F)$ is a Vector space.

e.g. $V(R)$: Real Vector Space.

$V(C)$: Complex Vector Space.

$V(Q)$: Rational Vector Space.

Note: In a vector space, We shall be dealing with two types of zero elements. One is zero vector and the other is zero element of the field i.e. Scalar.

Q. A field can be regarded as a vector space over any of its subfield.

Q. C is a Vector space over R , but R is not a vector space over C ?

GENERAL PROPERTIES OF VECTOR SPACE

- (i) $a\bar{0} = \bar{0} \quad \forall a \in F$
- (ii) $0\alpha = \bar{0} \quad \forall \alpha \in V$
- (iii) $a(-\alpha) = -(a\alpha)$
- (iv) $(-a)\alpha = -(a\alpha)$
- (v) $a(\alpha - \beta) = a\alpha - a\beta \quad ; \beta \in V$
- (vi) $a\alpha = \bar{0} \Rightarrow a = 0 \text{ or } \alpha = \bar{0}$



VECTOR SUBSPACE

Let V be a Vector space over the field F and let $W \subseteq V$. Then W is called a subspace of V if W itself is a VS over F with respect the operations of vector additions and scalar multiplication.

Theorem 1:

The necessary and sufficient condition for a non-empty subset W of a vector space $V(F)$ to be a subspace of V is the W is closed under vector addition and scalar multiplication in V .

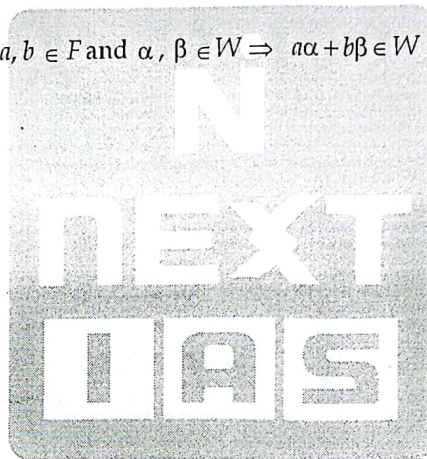
Theorem 2:

The N and S condition for a non empty subset W of a VS $V(F)$ to be a subspace of V are

- (i) $\alpha \in W, \beta \in W \Rightarrow \alpha - \beta \in W$
- (ii) $a \in F, \alpha \in W \Rightarrow a\alpha \in W$

Theorem 3:

The NSC for W to be a ss of V is $a, b \in F$ and $\alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W$.



ALGEBRA OF SUBSPACES

Theorem 1:

The intersection of any two subspaces W_1 and W_2 of a vector space $V(F)$ is also a subspace of $V(F)$.



Theorem 2.

The union of two subspaces is a subspace if and only if one is contained in the other.

**Smallest subspace containing any subset of $V(F)$**

Let $V(F)$ be a vector space and S be any subset of V . If U is a subspace of V containing S and is itself contained in every subspace of V containing S then U is called smallest SS containing S . The smallest subspace of V containing S is also called the subspace of V , generated or spanned by ' S ' and we shall denote it by the symbol $\{S\}$.

If $\{S\} = V$, then we say that V is spanned by S .

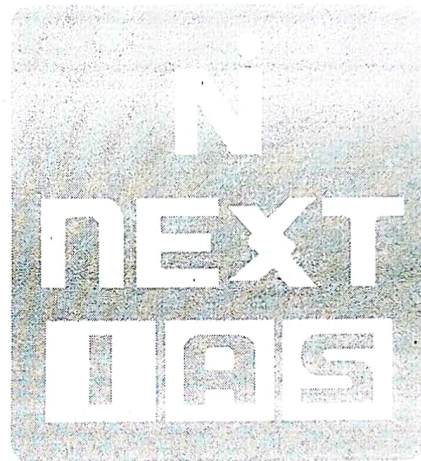
Linear combination

Let $V(F)$ be a vector space. If $\alpha_1, \alpha_2, \dots, \alpha_n \in V$, then any vector

$\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$, where $a_1, a_2, \dots, a_n \in F$ is called a linear combination of the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$.

Linear span

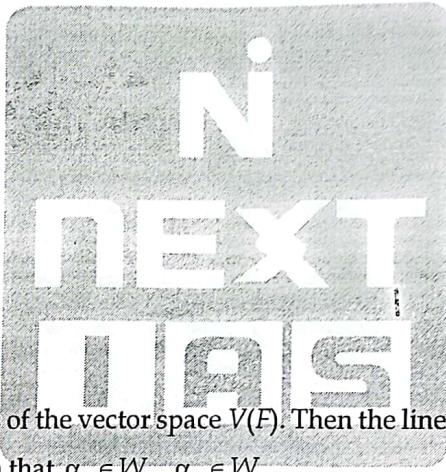
Let $V(F)$ be a vector space and S be any non empty subset of V . Then the linear span of S is the set of all linear combinations of finite sets of elements of S and is denoted by $L(S)$. Thus, we have



$L(S) = \{ a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n : \alpha_1, \alpha_2, \dots, \alpha_n \text{ is any arbitrary finite subset of } S \text{ and } a_1, a_2, \dots, a_n \text{ is any finite subset of } F \}.$

Theorem

The linear span $L(S)$ of any subset S of a vector space $V(F)$ is a subspace of V generated by S i.e. $L(S) = \langle S \rangle$.

**Linear sum of Two subspaces:**

Let W_1 and W_2 be two subspaces of the vector space $V(F)$. Then the linear sum of the subspaces W_1 and W_2 is the set of all sum $\alpha_1 + \alpha_2$ such that $\alpha_1 \in W_1, \alpha_2 \in W_2$.

Thus, $W_1 + W_2 = \{ \alpha_1 + \alpha_2 : \alpha_1 \in W_1 \text{ \& } \alpha_2 \in W_2 \}$

Theorem :

If W_1 and W_2 are subspaces of the vector space $V(F)$, then

- (i) $W_1 + W_2$ is a subspace of $V(F)$
- (ii) $W_1 + W_2 = \langle W_1 \cup W_2 \rangle$ i.e. $L(W_1 \cup W_2) = W_1 + W_2$

LINEAR DEPENDENCE AND LINEAR INDEPENDENCE OF VECTORS**Linear Dependence:**

Let $V(F)$ be a vector space, A finite set $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of vectors of V is said to be linearly dependent if there exist scalars $a_1, a_2, \dots, a_n \in F$ not all of them 0, such that $a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = 0$

Linear Independent :

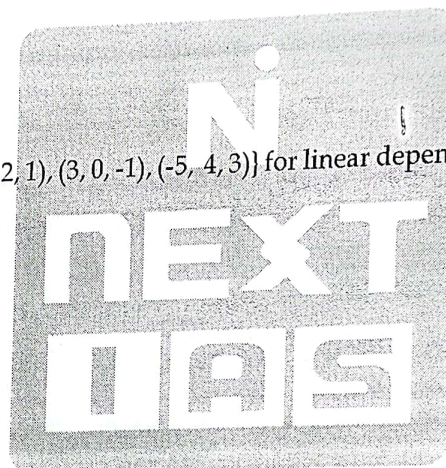
If every relation of the type/ from

$$a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = 0$$

$$\Rightarrow a_i = 0 \text{ for each } 1 \leq i \leq n$$

Q. Prove that if two vectors are linearly dependent, one of them is a scalar multiple of the other.

Q. Examine the set of vectors $\{(-1, 2, 1), (3, 0, -1), (-5, 4, 3)\}$ for linear dependence.



Some theorems :

1. Let $V(F)$ be a vector space $\alpha_1, \alpha_2, \dots, \alpha_n$ are non zero vectors $\in V$ then either they are linearly independent or some $\alpha_k, 2 \leq k \leq n$, is a linear combination of the preceding ones $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{k-1}$.

2. The set of non zero vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ of $V(F)$ is linearly dependent if some α_k , $2 \leq k \leq n$ is a linear combination of the preceding ones.



3. If in a vector space $V(F)$, a vector β is a linear combination of the set of vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ then the set of vectors $\beta, \alpha_1, \alpha_2, \dots, \alpha_n$ is linearly dependent.

4. Let S be a linearly independent subset of a vector space V . Suppose β is a vector in V which is not in the subspace spanned by S . Then the set obtained by adjoining β to S is linearly independent.



BASIS OF A VECTOR SPACE

Basis : A subset S of a vector space $V(F)$ is said to be a bases of $V(F)$, if

- (i) S consists of linearly independent vectors.
- (ii) S generates $V(F)$ i.e. $L(S) = V$ i.e. each vector in V is a linear combination of a finite number of elements of S .

Examples :

A system S consisting of n vector $e_1 = (1, 0, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $e_n, \dots, e_n = (0, 0, \dots, 1)$ is a basis of $V_n(F)$.

[This particular basis is called the standard basis of $V_n(F)$ or F^n].



Finite Dimensional Vector Space (FDVS)

The vector space $V(F)$ is said to be finite dimensional or finitely generated if there exists a finite subset S of V such that $V = L(S)$

Example :

- (i) The vector space $V_n(F)$ of n tuples is a FDVS.
- (ii) The vector space $F[x]$ of all polynomials over a field F is not finite dimensional (why !!)

Note : A VS which is not finitely generated may be referred to as an infinite dimensional space..

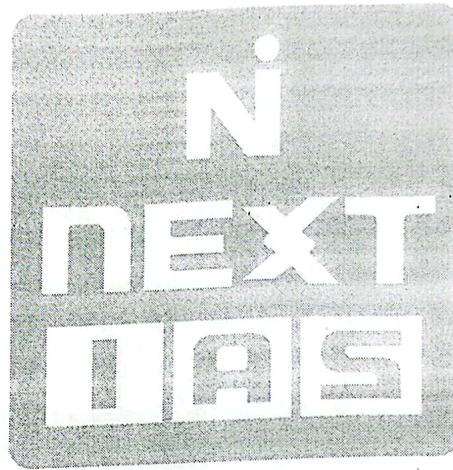
Theorem :

There exists a basis for each finite dimensional vector space.



Dimension Theorem of Vector space :

If $V(F)$ is a finite dimensional vector space, then any two bases of V have the same no. of elements.



Note: The number of elements in any basis of a FDVS $V(F)$ is called the dimension of the vector space $V(F)$ and will be denoted by $\dim V$.

Theorem :

(Extension Theorem) Every linearly independent subset of a finitely generated vector space $V(F)$ is either a basis of V or can be extended to form a basis.



Theorem :

Each set of $(n + 1)$ or more vectors of a finite dimensional vector space $V(F)$ of dimension n is linearly dependent.



Theorem :

Each subspace W of a $FDVS V(F)$ of dimension n is $FDVS$ with n is $m \leq n$.



Q. Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a basis of a FDVS $V(F)$ of dimension n . Then every element α of V can be uniquely expressed as $\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$

Where $a_1, a_2, \dots, a_n \in F$

Theorem :

If W_1, W_2 be two subspaces of a FDVS $V(F)$, then $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$.



HOMOMORPHISM OF VECTOR SPACE OR LINEAR TRANSFORMATION**Definition :**

Let $U(F)$ and $V(F)$ be two vector spaces. Then a mapping $f : U \rightarrow V$ is called homomorphism of U into V if

$$(i) \quad f(\alpha + \beta) = f(\alpha) + f(\beta) \quad \forall \alpha, \beta \in U \text{ and}$$

$$(ii) \quad f(a\beta) = a f(\beta) \quad \forall a \in F, \beta \in U$$

$$\text{i.e.} \quad f(a\alpha + b\beta) = a f(\alpha) + b f(\beta)$$

$$\forall a, b \in F \text{ and } \alpha, \beta \in U$$

If f is onto then, V is called Homomorphic image of U .

Note : If f is a homomorphism of $U(F)$ into $V(F)$, then

$$(i) \quad f(\bar{0}) = \bar{0}' \text{ where } \bar{0} \text{ and } \bar{0}' \text{ are the zero vectors of } U \text{ and } V \text{ respectively.}$$

$$(ii) \quad f(-\alpha) = -f(\alpha) \quad \forall \alpha \in U$$



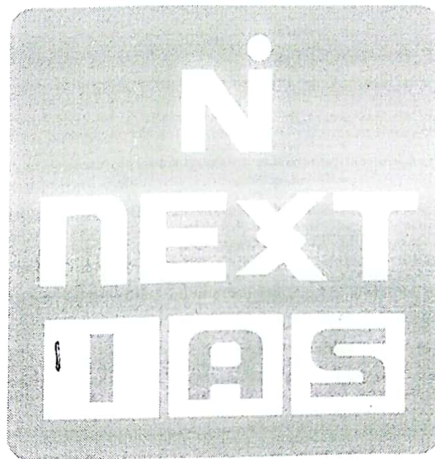
ISOMORPHISM OF VECTOR SPACES

Definition:

Let $U(F)$ and $V(F)$ be two vector spaces. Then a mapping $f:U \rightarrow V$ is called an isomorphism of U onto V if

- (i) f is one
- (ii) f is onto
- (iii) $f(a\alpha + b\beta) = af(\alpha) + bf(\beta) \forall a, b \in F, \alpha, \beta \in U$

Also, then the two vector spaces U and V are said to be isomorphic and symbolically we write $U(F) \cong V(F)$

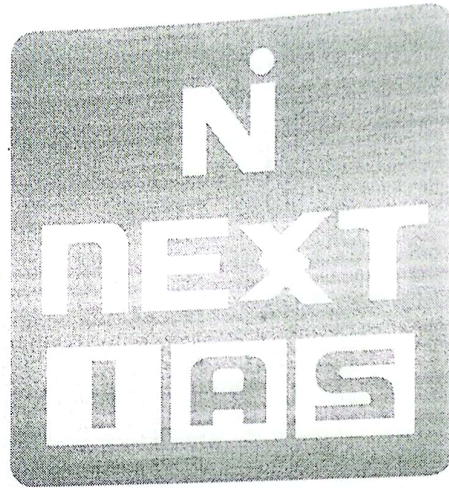


ISOMORPHISM OF $FDVS$

1. Two $FDVS$ over the same field are isomorphic iff if they are of the same dimension.



2. Every n dimensional vector space $V(F)$ is isomorphic to $V_n(F)$.



Coordinates:

Let $V(F)$ be a FDVS. Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be an ordered basis for V . By an ordered basis we mean that the vectors of B have been enumerated in some well defined way i.e. the vectors occupying the first, second, ..., n^{th} places in the set B are fixed.

Let $\alpha \in V$. Then there exists a unique n tuples $(x_1, x_2, \dots, x_n)$ of scalars such that

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \dots + x_n\alpha_n = \sum_{i=1}^n x_i \alpha_i$$

Then n tuples $(x_1, x_2, \dots, x_n)$ is called the n tuple of coordinates of α relative to the ordered basis B . The $n \times 1$

matrix $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ is called coordinate matrix of α relative to the ordered basis B .

Q. Find the Coordinates of the vector $(2, 1, -6)$ of R^3 relative to the basis $\alpha_1 = (1, 1, -2)$,

$$\alpha_2 = (3, -1, 0), \alpha_3 = (2, 0, -1)$$

Solution:

To find the coordinates of the vector $(2, 1, -6)$ relative to the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$ we shall express the vector

$(2, 1, -6)$ as a linear combination of the vectors $\alpha_1, \alpha_2, \alpha_3$

Let p, q, r be scalars in R such that $(2, 1, -6) = p\alpha_1 + q\alpha_2 + r\alpha_3$

$$\Rightarrow (2, 1, -6) = p(1, 1, -2) + q(3, -1, 0) + r(2, 0, -1)$$

$$\Rightarrow p + 3q + 2r = 2$$

$$p - q = 1$$

$$2p - r = -6$$

$$\text{On solving, we get } p = \frac{-7}{8}; q = \frac{-15}{8}; r = \frac{17}{4}$$

Hence the required coordinates

LINEAR TRANSFORMATION OR VECTOR SPACE HOMOMORPHISM**Define :**

Let $U(F)$ and $V(F)$ be two vector spaces over the same field F . A linear transformation from U into V is a function T from U into V such that $T(a\alpha + b\beta) = aT(\alpha) + bT(\beta)$ for all α, β in U and for all a, b in F .

Linear operator :

A linear operator on V is a function T from V into V such that $T(a\alpha + b\beta) = aT(\alpha) + bT(\beta)$ for all α, β in V and for all a, b in F .

Ex. $T: V_3(R) \rightarrow V_2(R)$, defined by $T(a, b, c) = (a, b) \forall a, b, c \in R$ is a LT from $V_3(R)$ into $V_2(R)$.

Note:

- (i) Zero Transformation: $T(\alpha) = \bar{0} \forall \alpha \in U$
- (ii) Identify operator: $T(\alpha) = \alpha \quad \forall \alpha \in U$
- (iii) Negative of linear transformation: $(-T)(\alpha) = -[T(\alpha)] \forall \alpha \in U$

Properties of LT: $T: U \rightarrow V$

- (i) $T(\bar{0}) = \bar{0}$, where $\bar{0}$ is zero vectors of U and $\bar{0}'$ is zero vector of V .
- (ii) $T(-\alpha) = -T(\alpha) \forall \alpha \in U$
- (iii) $T(\alpha - \beta) = T(\alpha) - T(\beta) \forall \alpha, \beta \in U$
- (iv) $T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n) = a_1T(\alpha_1) + a_2T(\alpha_2) + \dots + a_nT(\alpha_n)$

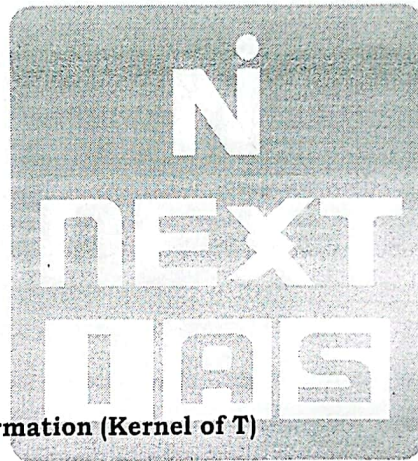
RANGE AND NULL SPACE OF LINEAR TRANSFORMATION

Range of a LT:

Let $U(F)$ and $V(F)$ be two vector spaces and let T be a linear transformation from U into V . Then the range of T written as $R(T)$ is the set of all vectors β in V such that $\beta = T(\alpha)$ for some α in U .

$$\text{Range}(T) = \{T(\alpha) \in V : \alpha \in U\}$$

Q. Let $T: U \rightarrow V, P.T \text{Range}(T)$ is a subspace of V .



Null Space of a linear Transformation (Kernel of T)

Null space of T written as $N(T)$ is the set of all vectors α in U such that $T(\alpha) = \bar{0}$.

$$\text{Thus, } N(T) = \{\alpha \in U : T(\alpha) = \bar{0} \in V\}$$

Q. Null space if T is a subspace of U , Prove.

Rank and Nullity of a Linear Transformation

- Rank of T denoted by $\rho(T)$ is the dimension of the range of T i.e.
 $\rho(T) = \dim R(T)$.
- Nullity of T denoted by $\nu(T)$ is the dimension of null space of T i.e.
 $\nu(T) = \dim N(T)$

Theorem:

Let U and V be vector spaces over the field F and let T be a linear transformation from U into V .

Then

$$\text{rank}(T) + \text{nullity}(T) = \dim U$$

{obviously U is finite dimensional}



Q. $T(x_1, x_2, x_3) = (x_1 - x_2 + 2x_3, 2x_1 + x_2 - x_3, -x_1 - 2x_2)$ Describe the null space of T .

Ans.

Zero sub space

Note:

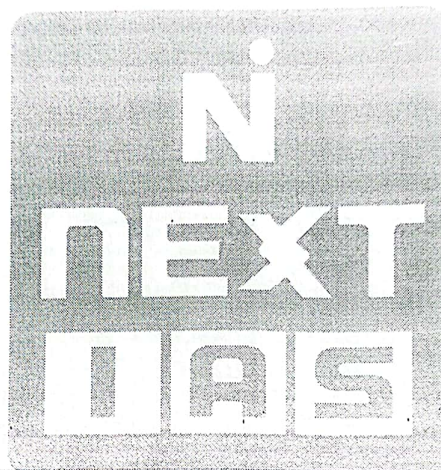
- Let T_1 and T_2 be LTs from U into V . The function $T_1 + T_2$ defined by $(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha) \forall \alpha \in U$ is a linear transformation from U into V . If c is any element of F , the function (cT) defined by $(cT)(\alpha) = cT(\alpha) \forall \alpha \in U$ is a LT from U into V .

The set $L(U, V)$ of all LTs from U into V together with the addition and scalar multiplication defined above is a vector space over the field F . Let U be an n dimensional vector space over the field F , and let V be an m -dimensional vector space over F . Then the vector space $L(U, V)$ of all linear transformations from U into V is finite dimensional and is of dimension mn .

Invertible Linear Transformation:

Let T be a linear transformation from U into V such that T is one one onto. Then T is called invertible.

Note : Let T be a linear transformation from U into V . If T is one one and onto, then the inverse function T^{-1} is a linear transformation from V into U .



SINGULAR AND NON SINGULAR TRANSFORMATIONS

1. T is said to non singular if the null space of T (i.e. $\text{Ker } T$) consist of the zero vector alone i.e. if $\alpha \in U$ and $T(\alpha) = \bar{0} \Rightarrow \alpha = \bar{0}$. If there exist a vector $\alpha \neq \bar{0} \in U$ s.t $T(\alpha) = \bar{0}$, then T is said to be singular

Note 1:

T is non singular iff T is one one.

Let T is non singular

Let, $\alpha_1, \alpha_2 \in U$

and $T(\alpha_1) = T(\alpha_2)$

$$\Rightarrow T(\alpha_1 - \alpha_2) = \bar{0}$$

$\therefore T$ is non singular

$$\Rightarrow \alpha_1 - \alpha_2 = \bar{0}$$

$$\Rightarrow \alpha_1 = \alpha_2$$

$\therefore T$ is 1-1.

Part II

Let T is one one.

$$\alpha \in U \text{ s.t. } T(\alpha) = \bar{0}$$

we know that $T(\bar{0}) = \bar{0}$

$$\Rightarrow T(\alpha) = T(\bar{0})$$

Since, T is 1-1.

$$\Rightarrow \alpha = \bar{0}$$

$\therefore T$ is non-singular.

2. Let U and V be finite dimensional vector spaces over the field F such that $\dim U = \dim V$. If T is a linear transformation from U into V , the following are equivalent.

- (i) T is invertible
- (ii) T is non singular
- (iii) The range of T is V .
- (iv) If $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is any basis for U , then $\{T(\alpha_1), \dots, T(\alpha_n)\}$ is a basis for V .

Q. Find a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(1, 0) = (1, 1)$ and $T(0, 1) = (-1, 2)$. Prove that T maps the square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$ into a parallelogram.

MATRIX REPRESENTATION OF LINEAR OPERATOR

Let T be a linear operator (transformation) from a vector space V into itself and suppose $S = \{u_1, u_2, \dots, u_n\}$ is a basis of V .

Now, $T(u_1), T(u_2), \dots, T(u_n)$ are vectors in V , and so each is a linear combination of the vectors in the basis S :

$$T(u_1) = a_{11}u_1 + a_{12}u_2 + \dots + a_{1n}u_n$$

$$T(u_2) = a_{21}u_1 + a_{22}u_2 + \dots + a_{2n}u_n$$

$\vdots$

$$T(u_n) = a_{n1}u_1 + a_{n2}u_2 + \dots + a_{nn}u_n$$

The transpose of the above matrix of coefficient denoted by $[T]_S$ is called the matrix representation of T relative to the basis S .

$[T]_S = [[T(u_1)], [T(u_2)], \dots, [T(u_n)]]$ i.e. the columns of $[T]_S$ are the coordinate vectors of $T(u_1), T(u_2), \dots, T(u_n)$ respectively.

Example :

Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator defined by $F(x, y) = (2x+3y, 4x-5y)$.

(a) Find the matrix representation of F relative to the basis $S = \{u_1, u_2\} = \{(1, 2), (2, 5)\}$

Solution.

First, find $F(u_1)$, and then write it as a linear combination of the basic vector u_1, u_2

$$F(u_1) = F\left[\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right] = \begin{bmatrix} 8 \\ -6 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$x + 2y = 8$$

$$\Rightarrow 2x + 5y = -6$$

$$x = 52, y = -22$$

$$F(u_1) = 52u_1 - 22u_2$$

Next, find $F(u_2)$, and then write it as a linear combination of u_1 and u_2 :

$$F(u_2) = F\left[\begin{pmatrix} 2 \\ 5 \end{pmatrix}\right] = \begin{bmatrix} 19 \\ -17 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$x + 2y = 19$$

$$2x + 5y = -17$$

$$x = 129, y = -55$$

$$\therefore F(u_2) = 129u_1 - 55u_2$$

Now write the coordinate of $F(u_1)$ and $F(u_2)$ as Columns to obtain the matrix.

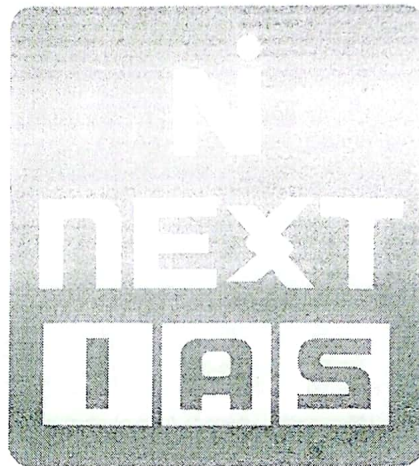
$$[F]_s = \begin{bmatrix} 52 & 129 \\ -22 & -55 \end{bmatrix}$$

B. Find the matrix representation of F relative to usual basis $E = \{e_1, e_2\} = \{1, 0\}, \{0, 1\}$

$$F(1, 0) = (2, 2) = 2e_1 + 4e_2$$

$$F(0, 1) = (3, -5) = 3e_1 - 5e_2$$

$$\text{and so, } [F]_r = \begin{bmatrix} 2 & 3 \\ 4 & -5 \end{bmatrix}$$



ALGORITHM FOR FINDING MATRIX REPRESENTATIONS

The input is a linear operator T on a vectors space V and a basis $\{u_1, u_2, \dots, u_n\}$ of V . The output is the matrix representation $[T]$.

Step 0:

Find a formula for the coordinates of an arbitrary vector V relative to the basis S .

Step 1:

Repeat for each basis vector u_k in S :

- (i) Find $T(u_k)$
- (ii) Write $T(u_k)$ as a linear combination of the basis vectors $u_1, u_2, \dots, u_n$.

Step 2:

Form the matrix $[T]_S$ whose columns are the coordinate vectors in step 1(b)

Change of basis:

"How do our representations (Matrix) change if we select another basis"

Definition :

Let $S = \{u_1, u_2, \dots, u_n\}$ be a basis of a vector space V , and let $S' = \{v_1, v_2, \dots, v_n\}$ be another basis. Because S is a basis; each vector in the new basis S' can be written uniquely as a linear combination of the vectors in S ; say

$$v_1 = a_{11}u_1 + a_{12}u_2 + \dots + a_{1n}u_n$$

$$v_2 = a_{21}u_1 + a_{22}u_2 + \dots + a_{2n}u_n$$

$$v_n = a_{n1}u_1 + a_{n2}u_2 + \dots + a_{nn}u_n$$

Let P be the transpose of the above matrix of coefficients; that is, let $P = [p_{ij}]$, where $p_{ij} = a_{ji}$. Then P is called the change of basis (or transition matrix) from the old basis S to the new basis S' .

Example:

Consider the following two base of R^2 ,

$$S = \{u_1, u_2\} = \{(1, 2), (3, 5)\} \text{ and}$$

$$S' = \{v_1, v_2\} = \{(1, -1), (1, -2)\}$$

(a) Find the change of basis matrix P from S to new basis S' .

Sol:

Write each of the new basis vectors of S' as linear combination of the original basis vectors u_1 and u_2 of S . we have

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ 5 \end{bmatrix} \Rightarrow x = -8, y = 3$$

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ 5 \end{bmatrix} \Rightarrow x = -11, y = 4$$

Thus $v_1 = 8u_1 + 3u_2$

$v_2 = -11u_1 + 4u_2$

and hence $P = \begin{bmatrix} -8 & -11 \\ 3 & 4 \end{bmatrix}$

(b) Find the change of basis matrix from the new basis S' to the old S .

Here, we write each of the old basis vector u_1 and u_2 of S as the linear combination of the new basis vector v_1 and v_2 of S' . This yields

$u_1 = 4v_1 - 3v_2$

$u_2 = 11v_1 - 8v_2$

and hence, $Q = \begin{bmatrix} 4 & 11 \\ -3 & -8 \end{bmatrix}$

kindly note $Q = P^{-1}$!! (as expected)

Note:

1. Change of basis matrix P from the usual/standard basis E to the basis $S = \{u_1, u_2, u_3\}$

Hence, we can directly write P :

$[u_1 \ u_2 \ u_3]$; Coordinates of u_1, u_2, u_3 appear as the columns in P .

e.g. $E = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

$S = \{(1, 0, 1), (2, 1, 2), (1, 2, 2)\}$

Here change of basis matrix from E to S will be $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 2 \end{bmatrix}$

$E \xrightarrow{P} S \xrightarrow{P^{-1}} E$

Theorem:

Let P be the change of basis matrix from a basis S to a basis S' in a vector space V . Then for any linear operator T on V ,

$$[T]_{S'} = P^{-1}[T]_S P$$

Theorem:

Two matrices represent the same linear operator if and only if the matrices are similar.

Theorem :

Let A be the matrix representation of a linear operator T . Then T is diagonalizable iff there exist an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

Matrix and General linear mappings

Suppose V and U are vector spaces over the same field K and $\dim V = m$ and $\dim U = n$. Further suppose $S = \{V_1, V_2, V_3, \dots, V_m\}$ and $S' = \{u_1, u_2, \dots, u_n\}$ are arbitrary but fixed bases of V and U respectively. Suppose $F: V \rightarrow U$ and

$$F(v_1) = a_{11}u_1 + a_{12}u_2 + \dots + a_{1n}u_n$$

$$F(v_2) = a_{21}u_1 + a_{22}u_2 + \dots + a_{2n}u_n$$

.....

$$F(v_m) = a_{m1}u_1 + a_{m2}u_2 + \dots + a_{mn}u_n$$

Then transpose of the above matrix of coefficient, denoted by $[F]_{ss'}$, is called the matrix representation of F relative to the bases S and S' .

Note:

1. Multiplying the coordinates of v in the basis S of V by $[F]$, we obtain the coordinates of $F(v)$.
2. Let S, S', S'' , be bases of vector spaces V, U, W respectively. Let $F: V \rightarrow U$ and $G: U \rightarrow W$ be linear mappings.

Then

$[GF]_{ss''} = [G]_{s's''} \cdot [F]_{s,s'}$, i.e. Matrix representation of the composition of two mappings is the matrix product of the matrix representation of the individual mappings.

3. Let P be change of basis matrix from a basis e to e' in V , and let Q be the change of basis matrix from a basis f to f' in U . Then for linear map $f: V \rightarrow U$,

$$[F]_{e'f'} = Q^{-1}[F]_{ef}P$$

i.e.

$$\begin{array}{ccc} e & & f \\ P \leftarrow \downarrow & & \downarrow \rightarrow Q \\ e' & & f' \end{array}$$

4. Let $F: V \rightarrow U$ be linear and $\text{rank}(F) = r$. Then there exist bases of V and U such that the matrix representation

$$\text{of } F \text{ has the form } A = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

Where I_r is the r -square identity matrix.

The above matrix A is called the normal or canonical form of the linear map F .

- Q. Let T be a linear operator on R^2 defined by :

$T(x, y) = (2y, 3x - y)$. find the matrix representation of T relative to the basis $\{(1, 3) (2, 5)\}$

Sol.

Let $\alpha_1 = (1, 3)$ and $\alpha_2 = (2, 5)$

$$\Rightarrow T(\alpha_1) = T(1, 3) = (6, 0)$$

$$T(\alpha_2) = T(2, 5) = (10, 1)$$

Now our aim is to express the vectors $T(\alpha_1)$ and $T(\alpha_2)$ as linear combination of the vectors in the basis $\{\alpha_1, \alpha_2\}$.

Let

$$\begin{aligned} (a, b) &= p\alpha_1 + q\alpha_2 \\ &= p(1, 3) + q(2, 5) \\ &= (p + 2q, 3p + 5q) \end{aligned}$$

Then,

$$\begin{aligned} p + 2q &= a \\ 3p + 5q &= b \end{aligned}$$

Solving these equations, we get

$$\begin{aligned} p &= -5a + 2b \\ q &= 3a - b \end{aligned}$$

Putting $a = 6, b = 0$ in (i) we get $p = -30, q = -18$

$$\therefore T(\alpha_1) = (6, 0) = -30\alpha_1 + 18\alpha_2 \quad \dots(i)$$

Again putting $a = 10, b = 1$ in (1) we get $p = -48, q = 29$

$$\therefore T(\alpha_2) = (10, 1) = -48\alpha_1 + 29\alpha_2 \quad \dots(ii)$$

From relation (i) and (ii), we see that the matrix of T relative to the basis $\{\alpha_1, \alpha_2\}$ is $\begin{bmatrix} -30 & -48 \\ 18 & 29 \end{bmatrix}$

MATRICES

Fundamentals :

- **The Transpose of a matrix :**

The matrix obtained from any given matrix A , by interchanging its rows and columns is called the Transpose of A and is usually denoted by A^T .

$$A^T = [b_{ij}]$$

where $b_{ij} = a_{ji}$

e.g.

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix}$$

$\Rightarrow$

$$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \end{bmatrix}$$

- **The conjugate of a matrix :**

The matrix obtained from any given matrix A , on replacing its elements by the corresponding conjugate complex numbers is called the conjugate of A and is denoted by $\bar{A}$.

$$A = \begin{bmatrix} 1+i & 1+2i \\ 2+3i & 2 \end{bmatrix}$$

$\Rightarrow$

$$\bar{A} = \begin{bmatrix} 1-i & 1-2i \\ 2-3i & 2 \end{bmatrix}$$

- **Conjugate Transpose of a matrix A :**

$$A^\theta = (\bar{A})^T$$

- **Triangular matrix**

A matrix $A = [a_{ij}]$ is said to be upper triangular if ;

$$i > j \Rightarrow a_{ij} = 0$$

also is said to be lower triangular if;

$$i < j \Rightarrow a_{ij} = 0$$

- **Symmetric matrix**

If $A = A^T$ then A is symmetric matrix.

- **Skew Symmetric matrix**

$$A^T = -A$$

i.e. (i, j) th element is the negative of its (j, i) th element.

Q. Every diagonal element of a skew Hermitian matrix is either a pure imaginary number or zero?

- Hermitian matrix:

$$A^0 = A$$

i.e. every (i, j) th element is the conjugate complex of the (j, i) th element.

Q. Every diagonal element of a skew Hermitian matrix is either a pure imaginary number or zero.

- Skew Hermitian matrix

$$A^0 = -A$$

- Minors of Matrix:

The determinant of every square submatrix of a matrix is called a minor of the matrix.



ALGEBRA OF MATRIX

Multiple of a matrix by numbers:

$\Rightarrow$ Let $k \in F$, then kA is the matrix obtained by multiplying every element of the matrix A by k .

- Q. (i) A is symmetric. $PTkA$ is symmetric.
 (ii) A is Hermitian. $PTiA$ is skew Hermitian
 (iii) A is skew-Hermitian. $PTiA$ is Hermitian.

Sum of Matrices:

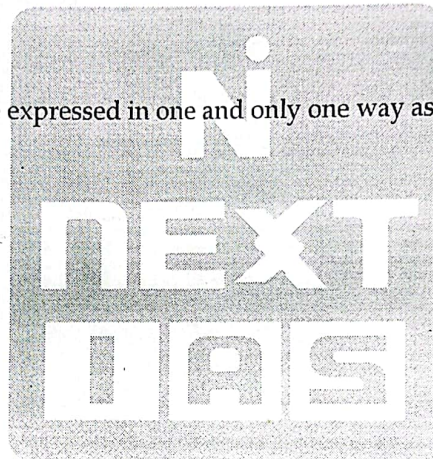
Let $A = [a_{ij}]$ and $B = [b_{ij}]$

then $C = A + B$ where $c_{ij} = a_{ij} + b_{ij}$

- Q. If $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & -4 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 4 & 3 \\ 3 & 9 & 8 \end{bmatrix}$, $C = \begin{bmatrix} 7 & 6 & -4 \\ 3 & 1 & -2 \end{bmatrix}$ evaluate $2A + 3B - 5C$.

Ans. $\begin{bmatrix} -27 & -12 & 39 \\ -2 & 14 & 52 \end{bmatrix}$

- Q. Every square matrix A can be expressed in one and only one way as $P + iQ$ where P and Q are Hermitian matrices.

**Product of Matrices:**

Let $A_{m \times n}$ and $B_{n \times p}$ be two matrices.

Then $C_{m \times p}$ such that $c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$ is the product AB of the matrices A and B taken in this order.

Note: The product AB has a meaning iff the number of columns of A equals that of rows of B .

$$AB = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_m \end{bmatrix} [C_1, C_2, \dots, C_p] = \begin{bmatrix} R_1 C_1, R_1 C_2, \dots, R_1 C_p \\ R_2 C_1, R_2 C_2, \dots, R_2 C_p \\ \vdots \\ R_m C_1, R_m C_2, \dots, R_m C_p \end{bmatrix}$$

Q. Let $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ compute AB and BA and show that $AB \neq BA$.

Reversal law for Transpose:

$$(AB)^T = B^T A^T$$

Also,

$$(AB)^0 = B^0 A^0$$

Associative law :

$$AB(C) = A(BC)$$

Q. $(A \pm B)^2 \neq A^2 \pm 2AB + B^2$

Q. $A^2 - B^2 \neq (A+B)(A-B)$

Distributive law :

$$A(B+C) \neq AB+AC$$

Zero Divisors:

If A and B are non zero matrices such that $AB = 0$, then A, B are called divisors of zero.

Note: Cancellation law does not hold for matrix multiplication.

i.e.

$$AB = AC \not\Rightarrow B = C$$

Some special type of matrices:

1. Idempotent matrix:

$$A^2 = A$$

2. Nilpotent matrix:

$$A^K = 0$$

where $K \in \mathbb{Z}^+$

If, however, K is the least positive integer for which $A^K = 0$, then K is the index of the nilpotent matrix A .

3. Involutory Matrix:

$$A^2 = I$$

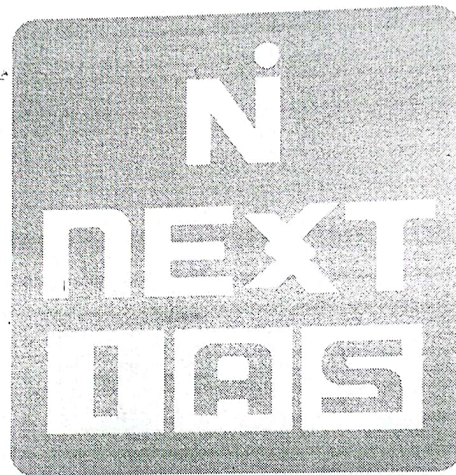
4. Orthogonal matrix:

$$AA^T = I$$

5. Unitary matrix:

$$AA^0 = I$$

6. Magic Square : Sum of elements of any row / Column / diagonal is same.



ADJOINT OF A SQUARE MATRIX

It is the transpose of the matrix formed by the co-factors of A .

e.g.

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -1 \\ 2 & 0 & -4 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 0 \\ 3 & -1 & -4 \end{bmatrix}$$

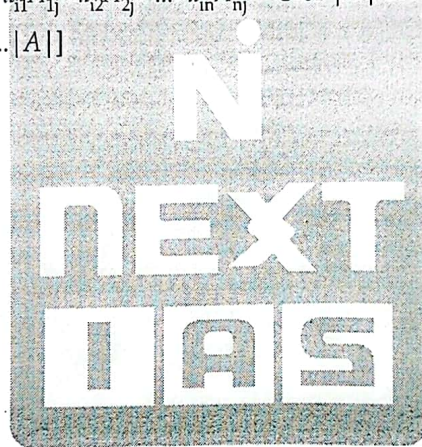
$$\text{Adj } A = \begin{bmatrix} -4 & 4 & -4 \\ -2 & -10 & 1 \\ -1 & 2 & 1 \end{bmatrix}$$

Relation between A and $\text{Adj } A$:

$$A(\text{adj } A) = |A|I = (\text{adj } A)A.$$

The (i, j) th element of $A \cdot \text{adj } A = a_{i1}A_{1j} + a_{i2}A_{2j} + \dots + a_{in}A_{nj} = 0$ or $|A|$ according as $j \neq i$ or $j = i$

Thus $A \cdot \text{adj } A = \text{Diag. } [|A| \ |A| \ \dots \ |A|]$



INVERSE OF MATRIX

Let, $AB = BA = I$

If there exist such B then it is called inverse of A .

- Non square matrices cannot possess inverse.
- The necessary and sufficient condition for A to possess inverse is $|A| \neq 0$.
- $(AB)^{-1} = B^{-1}A^{-1}$ (Reversal law)
- $(A^T)^{-1} = (A^{-1})^T$
- $A^{-1} = \frac{\text{adj}A}{|A|}$

Trace of a matrix:

Let A be a square matrix of order n . The sum of the elements of A lying along the principal diagonal is called the trace of A .

$$\begin{aligned} \text{tr. of } A &= \sum_{i=1}^n a_{ii} \\ &= a_{11} + a_{22} + \dots + a_{nn} \end{aligned}$$

Prove that:

- (i) $\text{tr}(\lambda A) = \lambda \text{tr} A$
- (ii) $\text{tr}(A + B) = \text{tr} A + \text{tr} B$
- (iii) $\text{tr}(AB) = \text{tr}(BA)$

DETERMINANTS

Important points :

- If two adjacent rows or columns of a determinant are interchanged, the sign of the determinant is changed, the numerical value remaining unaltered.
- The determinants of matrix A and A^T are same.
- If every element in any (or column) be multiplied by the same factor, the determinant is multiplied by that factor.
 $\Rightarrow |kA| = k^n |A|$
- If two rows or columns are identical the value of the determinant is zero.
- If each constituent of any row or any column be the sum (difference) of two quantities, the determinant can be expressed as the sum (different) of two determinants of the same order.
- The value of determinant remains unchanged if all the elements of one row (column) are multiplied by a scalar and added to the corresponding elements of a different row (column).

The Cofactor of an element:

In the expansion of a determinant A , we may collect all terms containing the fixed element a_{ij} as a factor and write their sum in factorised form as $a_{ij}A_{ij}$, where A_{ij} is the factor remaining when the element a_{ij} is factored out. A_{ij} is called the co factor of the element a_{ij} is determinant A .

$$\det. A = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}$$

$$= \sum_{k=1}^n a_{kj}A_{kj}$$

$$(j = 1, 2, \dots, n)$$

- Let $A = (a_{ij})$ be a matrix of order n . Then if $i \neq j$, we have

$$= \sum_{k=1}^n a_{ij}A_{jk} = 0 \text{ and } \sum_{k=1}^n a_{ki}A_{kj} = 0$$

This property says that sum of the products of the elements of any row (column) by the cofactor of the corresponding elements of a different row (column) is always zero.

Q.
$$\begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ca \\ ab & b^2+bc & c^2 \end{vmatrix} = 4a^2b^2c^2$$

- Q. A skew symmetric determinant of odd order vanishes.

USE OF DETERMINANTS IN SOLVING LINEAR EQUATIONS

Cramer's Rule:

Consider the equations:

$$a_1x + b_1y + c_1z = m_1$$

$$a_2x + b_2y + c_2z = m_2$$

$$a_3x + b_3y + c_3z = m_3$$

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$x = \frac{\begin{vmatrix} m_1 & b_1 & c_1 \\ m_2 & b_2 & c_2 \\ m_3 & b_3 & c_3 \end{vmatrix}}{\Delta}$$

$$y = \frac{\begin{vmatrix} a_1 & m_1 & c_1 \\ a_2 & m_2 & c_2 \\ a_3 & m_3 & c_3 \end{vmatrix}}{\Delta}$$

$$z = \frac{\begin{vmatrix} a_1 & b_1 & m_1 \\ a_2 & b_2 & m_2 \\ a_3 & b_3 & m_3 \end{vmatrix}}{\Delta}$$

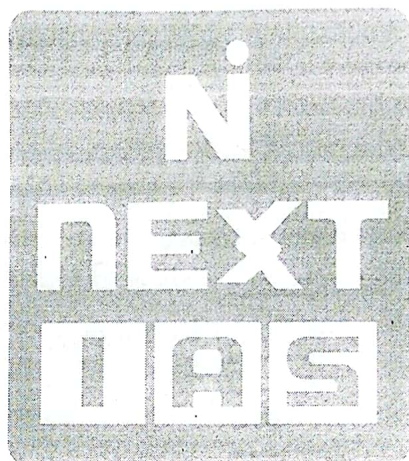


RANK OF MATRIX

Definition:

The number ' r ' with the following two properties is called the rank of the matrix:

- (i) There is at least one non zero minor of order r .



(ii) Every minor of order $(r+1)$ is zero.

Rank is denoted by $\rho(A)$

e.g., determine the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 2 & 2 \end{bmatrix}$

→ Rank of every n rowed non singular matrix A is n .

→ $r(A) = \rho(A^T) = \rho(A^0)$

Nullity of Matrix:

If A is a square matrix of order, n then $n - \rho(A)$ is called the nullity of the matrix A and is denoted by $N(A)$.

Nullity of non singular matrix is 0.

Elementary transformations of a Matrix:

1. Interchange of two rows (columns)
2. The multiplication of the elements of a row (column) by a non zero number.
3. The addition to the elements of a row (column), the corresponding elements of a row (column) multiplied by any number.

Statements:

1. Interchange of a pair of rows does not alter the rank.
2. Multiplication of the elements of row by a non zero number does not alter the rank.
[i.e., Elementary transformations of a matrix do not alter its rank]

Q. Find the rank of the matrices

(i) $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & -2 & -1 & -7 \\ 0 & 2 & 1 & -5 \\ 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & 6 \end{bmatrix}$

Note:

1. The rank of a matrix does not alter on multiplication with a non singular matrix.
2. The rank of the product of two matrices cannot exceed the rank of either matrix.
i.e., $\rho(AB) \leq r(A)$ and $\rho(AB) \leq \rho(B)$

Finding inverse of a non singular matrix by elementary row transformations

Step-1: Which $A = IA$

Step-2: Go on performing suitable elementary row transformations on the matrix A on the left and the pre-factor of A on the right till we reach the result $I = BA$

Then B is inverse of A .

Q. Compute the inverse of $A = \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 1 & 0 \end{bmatrix}$

Sol. $\begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Performing $R_{21}(1), R_{31}(2), R_{41}(-1)$

$$\begin{bmatrix} -1 & -3 & 3 & -1 \\ 0 & -2 & 2 & -1 \\ 0 & -11 & 8 & -5 \\ 0 & 4 & -3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} A$$

Performing $R_{32}\left(-\frac{11}{2}\right), R_{42}(2)$

$$\begin{bmatrix} -1 & -3 & 3 & -1 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -3 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -\frac{7}{2} & \frac{11}{2} & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} A$$

Performing $R_{43}\left(\frac{1}{3}\right)$

$$\begin{bmatrix} -1 & -3 & 3 & -1 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -3 & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -\frac{7}{2} & \frac{11}{2} & 1 & 0 \\ -\frac{1}{4} & \frac{1}{6} & \frac{1}{8} & 1 \end{bmatrix} A$$

Performing $R_{34}(-3), R_{24}(6), R_{14}(16)$

$$\begin{bmatrix} -1 & -3 & 3 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 1 & \frac{1}{9} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & 6 \\ 0 & 2 & 2 & 6 \\ -3 & -6 & 0 & -3 \\ -\frac{1}{4} & \frac{1}{6} & \frac{1}{3} & 1 \end{bmatrix} A$$

Performing $R_{23}\left(\frac{2}{3}\right), R_{13}(1)$

$$\begin{bmatrix} -1 & -3 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & \frac{1}{6} \end{bmatrix} = \begin{bmatrix} -3 & -5 & 2 & 2 \\ -2 & -2 & 2 & 4 \\ -3 & -6 & 0 & -3 \\ -\frac{1}{4} & \frac{1}{6} & \frac{1}{3} & 1 \end{bmatrix} A$$

Performing $R_{12}\left(-\frac{3}{2}\right)$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & \frac{1}{9} \end{bmatrix} = \begin{bmatrix} 0 & -2 & -1 & -3 \\ -2 & -2 & 2 & 4 \\ -3 & -6 & 0 & -3 \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{3} & 1 \end{bmatrix} A$$

Finally performing $R_4(-1), R_2\left(-\frac{1}{2}\right), R_3\left(-\frac{1}{3}\right), R_4(6)$

$$I = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix} A$$

Equivalence of Matrices:

$A_{m \times n}$ and $B_{m \times n}$ are equivalent if there exists two non singular matrices P, Q whose elements are members of F such that

$$A = PBQ$$

Criterion: Two $m \times n$ matrices over a field F are equivalent, iff they have the same rank.

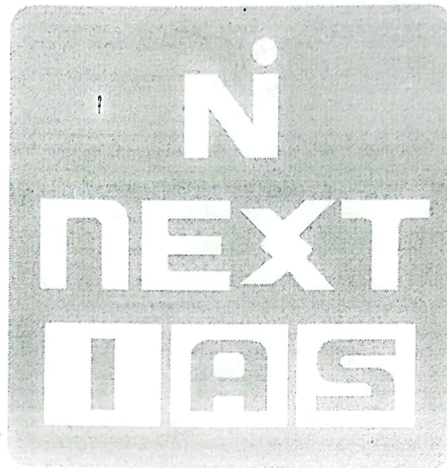
Row space of $m \times n$ matrix A : The space spanned by the m rows which is a subspace of V_n is called row space of A .

Column space: The space spanned by n columns which is a subspace of V_m is called column space.

Note:

1. The row rank is same as rank.
2. The column rank if a matrix is same as its rank.
3. The rank of matrix is equal to the maximum number of its linearly independent rows and also to the maximum number of its linearly independent columns.
4. $\rho(A+B) \leq \rho(A) + \rho(B)$
5. $\min. [\rho(B), \rho(A)] \geq \rho(AB) \geq \rho(A) + \rho(B) - n$

6. Every real skew symmetric matrix has an even rank.



Nature of Characteristics Roots of Some Special Types of Matrices:

1. The characteristic root of a Hermitian matrices are all real.

Let λ be characteristic root of a Hermitian matrix A so that there exists a vector $X \neq 0$, such that $AX = \lambda X$.

Taking conjugate transpose:

$$(AX)^0 = (\lambda X)^0$$

$$X^0 A^0 = \bar{\lambda} X^0$$

$$X^0 A = \bar{\lambda} X^0$$

$$(\because A^0 = A)$$

$$X^0 AX = \bar{\lambda} X^0 X$$

(Post multiply by X)

$$X^0(\lambda X) = \bar{\lambda} X^0 X$$

$$\Rightarrow X^0 X = \bar{\lambda} X^0 X$$

$$\Rightarrow X^0 X = \bar{\lambda} X^0 X$$

$$\Rightarrow \lambda = \bar{\lambda}$$

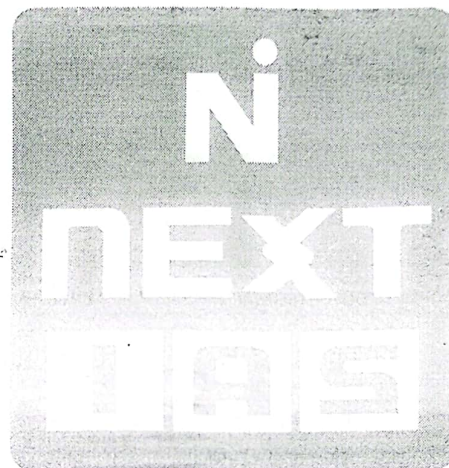
$\Rightarrow \lambda$ is real.

2. The characteristics roots of a real symmetric matrix are all real, for every such matrix is Hermitian.
3. A characteristic root of a skew Hermitian matrix is either zero or a pure imaginary number.
4. The modulus of characteristics root of a unitary matrix is unity.

Algebraic and Geometric Multiplicity of a CH. Root

If λ be a t -ple root of the ch. equation $|A - xI| = 0$ then, t , is called the algebraic multiplicity of λ and the dimension, s , of the characteristic space of A corresponding to λ , i.e., the number of linearly independent solutions of $(A - \lambda I)X = 0$ is called geometric multiplicity of A .

- Note:** 1. The characteristic roots of a diagonal matrix are the same as its diagonal elements.
2. Geometric multiplicity $\leq$ Algebraic multiplicity



Theorem 1:

Any system of characteristic vectors $X_1, \dots, X_k$ corresponding respectively to a system of distinct characteristic roots $\lambda_1, \lambda_2, \dots, \lambda_k$ of a matrix is linearly independent..



Theorem 2:

Any two characteristic vectors corresponding to two distinct characteristic roots of a Hermitian matrix are orthogonal.



Theorem 3:

Any two characteristic vectors corresponding to two distinct characteristic roots of a unitary matrix are orthogonal.



SIMILARITY OF MATRICES

A matrix 'B' over a field F , is said to be similar to a matrix 'A' over F , if there exists a nonsingular matrix P over F such that $B = P^{-1}AP$

Q. Show that the rank of every matrix similar to A is the same as that of A .

Theorem-1:

The necessary and sufficient condition for an n rowed matrix A to be similar to a diagonal matrix is that the set of characteristic vectors of A includes a set of ' n ' linearly independent vectors.



Theorem-2:

The necessary and sufficient condition for a square matrix to be similar to a diagonal matrix is that the geometric multiplicity of each of its characteristic roots coincides with the algebraic multiplicity.

Note: Every matrix with distinct characteristic roots is similar to a diagonal matrix.



Unitary similarity: A matrix B is said to be unitarily similar to a matrix A , if there exists a unitary matrix P such that $B = P^{-1}AP$

Normal matrices: A matrix is normal if $AA^0 = A^0A$



CONGRUENCE OF MATRICES

An n rowed square matrix B over F is said to be congruent to another n rowed square matrix A over F , if there exists a non singular matrix P over F such that $B = P^T A P$.

